Unbiased Estimators & Confidence Intervals

ANSWER KEY

Section 1: Unbiased Estimator of Population Mean & Variance

1 Given sample observations $x_1, \ldots, x_n$, what is the formula for the unbiased estimator s^2_{n-1} of the population variance σ^2?

a. $s^2_{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ ✓

b. $s^2_n = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum_{i=1}^{n}x_i^2$

d. $s^2 = \frac{(\sum x_i)^2}{n-1}$

Dividing by $n-1$ removes the downward bias of the sample variance, ensuring that $E(s^2_{n-1}) = \sigma^2$.

2 A sample of $n = 5$ values has $\sum x_i = 30$ and $\sum x_i^2 = 210$. Calculate the unbiased estimate of the population variance s^2_{n-1}.

a. 7.5 ✓

b. 6.0

c. 30

d. 9.0

$\bar{x} = 30/5 = 6$. $\sum(x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = 210 - 5(36) = 210 - 180 = 30$. Unbiased variance: $s^2_{n-1} = \frac{30}{5-1} = \frac{30}{4} = 7.5$.

3 If a sample standard deviation computed with denominator n is $s_n = 4.0$ for $n = 16$, what is the unbiased standard deviation s_{n-1}?

a. $s_{n-1} = 4.0 \times \sqrt{\frac{16}{15}} \approx 4.13$ ✓

b. $s_{n-1} = 4.0 \times \sqrt{\frac{15}{16}} \approx 3.87$

c. $s_{n-1} = 4.0$

d. $s_{n-1} = 4.0 \times \frac{16}{15} \approx 4.27$

$s^2_{n-1} = \frac{n}{n-1}s^2_n = \frac{16}{15}(16) \Rightarrow s_{n-1} = 4\sqrt{\frac{16}{15}} \approx 4.13$.

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Section 2: Confidence Intervals for Mean (Known Variance)

4 A random sample of $n = 64$ items from a population with known $\sigma = 12$ has mean $\bar{x} = 50$. What is the 95% confidence interval for μ ($z^* = 1.96$)?

a. $[47.06, 52.94]$ ✓

b. $[48.50, 51.50]$

c. $[44.12, 55.88]$

d. $[49.63, 50.37]$

Standard error is $\frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{64}} = \frac{12}{8} = 1.5$. Margin of error is $1.96(1.5) = 2.94$. Interval: $50 \pm 2.94 = [47.06, 52.94]$.

5 To reduce the margin of error of a confidence interval by a factor of 3, the sample size must be multiplied by:

a. 9 ✓

b. 3

c. $\sqrt{3}$

d. 6

Since margin of error is proportional to $\frac{1}{\sqrt{n}}$, to reduce it by 3 requires $\sqrt{n_{\text{new}}} = 3\sqrt{n} \Rightarrow n_{\text{new}} = 9n$.

6 What is the critical value z^* for a 99% confidence interval?

a. 2.576 ✓

b. 1.960

c. 1.645

d. 2.326

For a 99% two-tailed interval, each tail contains 0.005 of the probability. The corresponding standard normal critical value is $z^* \approx 2.576$.

Section 3: Confidence Intervals for Mean (Unknown Variance, t-distribution)

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7 When constructing a confidence interval for μ with unknown population variance σ^2 from a normal sample of size $n = 10$, which distribution is used?

a. **Student's t-distribution with 9 degrees of freedom ✓**

b. Standard Normal Z-distribution

c. Student's t-distribution with 10 degrees of freedom

d. Chi-square distribution with 9 degrees of freedom

Replacing σ with sample standard deviation s_{n-1} necessitates the t-distribution with degrees of freedom $df = n - 1 = 10 - 1 = 9$.

8 How does the critical value t^* compare to z^* for the same confidence level and finite sample size?

a. **$t^* > z^*$, because the t-distribution has heavier tails to reflect estimation uncertainty ✓**

b. $t^* < z^*$, because t is more compact

c. $t^* = z^*$ always

d. t^* can be smaller or larger depending on $\bar{x}$

The t-distribution exhibits greater tail dispersion than the standard normal distribution to account for estimating σ from sample data, making $t^* > z^*$ for any finite degrees of freedom.

9 A sample of $n = 16$ has $\bar{x} = 22.0$ and $s_{n-1} = 2.0$. Given $t_{0.025,\,15} = 2.131$, find the 95% confidence interval for μ.

a. $[20.93, 23.07]$ ✓

b. $[21.47, 22.53]$

c. $[19.87, 24.13]$

d. $[20.00, 24.00]$

Standard error $= \frac{s}{\sqrt{n}} = \frac{2.0}{4} = 0.50$. Margin of error $= 2.131(0.50) = 1.0655$. Interval: $22.0 \pm 1.07 = [20.93, 23.07]$.

Section 4: Interpretation of Confidence Levels

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10 Which statement correctly conveys the meaning of a '95% confidence level'?

a. If many random samples are drawn and intervals constructed, 95% of them will contain the true population parameter ✓

b. There is a 95% probability that the true parameter lies inside the specific interval $[47, 53]$

c. 95% of sample data values lie between 47 and 53

d. The sample mean has a 95% chance of being correct

The confidence level describes the long-run coverage rate of the interval estimation process over repeated random sampling.

11 If a 95% confidence interval for μ is $[102, 118]$, what is the conclusion for the two-tailed test $H_0 : \mu = 100$ vs $H_1 : \mu \neq 100$ at $\alpha = 0.05$?

a. Reject H_0 at the 5% level, because 100 lies outside the 95% confidence interval ✓

b. Fail to reject H_0, because 100 is close to 102

c. Accept H_0 as proven

d. The test is inconclusive without a P-value

A hypothesized value outside a $(1 - \alpha)$ confidence interval is rejected at significance level α in a two-tailed test.

12 What happens to the width of a confidence interval if the confidence level is increased from 90% to 99% (with sample size fixed)?

a. The interval widens ✓

b. The interval narrows

c. The interval width remains identical

d. The interval shifts to the right

Higher confidence requires capturing a larger portion of the sampling distribution, which increases the critical value and widens the margin of error.

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